SEUNG-WA AKPAN

Python & Backend Engineer · AI / LLM Systems

seungwaakpan@gmail.com · Nigeria — WAT (UTC+1) · Open to fully remote

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PROFESSIONAL PROFILE

First Class Computer Science graduate and backend engineer with 3+ years building production Python services and AI-powered platforms across startup, fellowship, and internship environments. Designs distributed processing architectures — Flask, Celery, Redis, and serverless GPU workers — for transcription, translation, and retrieval-augmented generation workloads, and ships well-tested, documented REST APIs on Linux and Docker. Equally productive in the Node.js and TypeScript ecosystem. Founded a university's first Google Developer Student Club chapter and mentors junior engineers.

CORE SKILLS & TECHNOLOGIES

Python & AI/ML
Python, FastAPI, Flask, Celery, pytest, LangChain, RAG pipelines, OpenAI, Google Gemini, OpenAI Whisper, M2M100, vector embeddings

Infrastructure & DevOps
Docker, AWS, Linux, Git, CI/CD pipelines, RunPod Serverless GPU, MinIO / S3

Databases & Messaging
PostgreSQL, MySQL, MongoDB, Redis, ChromaDB, BullMQ

APIs & Security
RESTful API design, WebSockets, JWT authentication, OAuth 2.0, rate limiting, input sanitisation, CORS

JavaScript / TypeScript
Node.js, TypeScript, NestJS, Express.js, React, Jest

Testing & Documentation
pytest, Jest, Swagger / OpenAPI, Postman

PROFESSIONAL EXPERIENCE

Python Developer · *Kevin Winter Foundation* | Remote | January 2025 – Present

- Built an end-to-end AI platform for video transcription, translation, and multi-format document export (PDF/DOCX), owning the service from architecture through production.
- Designed a distributed processing architecture using Flask, Celery, Redis, and RunPod serverless GPU workers to run AI workloads that exceed single-machine capacity.
- Integrated OpenAI Whisper for multilingual speech-to-text and Meta's M2M100 model for direct many-to-many translation.
- Developed a React and TypeScript front end with real-time job progress streamed over WebSockets.

Backend Developer · *KSolutions* | Nigeria | February 2025 – March 2026

National Youth Service Corps — Nigeria's mandatory one-year graduate service placement

- Developed the backend for a people-matching application using Node.js, Express, and SQL, covering appointment scheduling and match status management.
- Designed and implemented user-matching algorithms with compatibility scoring, plus an admin dashboard API for match coordinators.
- Resolved critical synchronisation defects between appointment status endpoints, restoring data consistency across the system.
- Applied Docker, Sequelize ORM, JWT authentication, and RESTful design patterns throughout.

Backend Engineering Intern · *HNG Tech* | Remote | October 2025 – December 2025

Competitive open-enrolment engineering fellowship — reached finalist stage

- Built production-ready APIs with Node.js, TypeScript, and NestJS for an AI-powered chat platform.
- Integrated Google Gemini to deliver context-aware conversations with persistent chat history in MongoDB.
- Implemented real-time messaging over Socket.IO with Redis-based rate limiting, tone personalisation, and S3-compatible (MinIO) file uploads.
- Stabilised staging and production by resolving database migrations, aligning API routes with DevOps infrastructure, and maintaining high test coverage.

Python Developer · *TIIDELAB* | Remote | June 2024 – August 2024

Pre-fellowship programme

- Built an AI-powered text-to-image conversion service integrating external AI APIs with multi-format image export.
- Optimised API response times by 40% through more efficient data processing and robust error handling.

Backend Developer Intern · *Resourceful Dev* | Abuja, Nigeria | July 2023 – September 2023

- Designed and built a Blog API in Node.js and Express with JWT authentication and role-based authorisation.
- Added a Redis caching layer that reduced database queries by 60%, and reached 85% test coverage with Jest.

SELECTED PROJECTS

Document RAG Service — *NestJS · TypeScript · ChromaDB · LangChain · OpenRouter · Docker*

github.com/seung-waedet/document-rag-service

- Retrieval-augmented generation API that ingests PDF, DOCX, and TXT files, chunks and embeds them into ChromaDB, and answers natural-language questions with an LLM.
- Used LangChain's RecursiveCharacterTextSplitter for document chunking and OpenRouter for provider-flexible model access (Meta Llama 3).
- Fully documented with Swagger and containerised via Docker Compose for local vector-database management.

Wallet Service — *NestJS · TypeScript · PostgreSQL · Paystack · Google OAuth · JWT*

github.com/seung-waedet/wallet-service

- Secure digital wallet API supporting registration, JWT and Google OAuth authentication, and full transaction management — deposits, withdrawals, and transfers.
- Integrated Paystack for payment processing and TypeORM for PostgreSQL schema management and migrations.
- Implemented bcrypt password hashing, SQL-injection prevention, and enforced HTTPS.

EDUCATION & CERTIFICATIONS

B.Sc. Computer Science — First Class Honours (CGPA 4.73 / 5.00) — Arthur Jarvis University, Cross River State, Nigeria | February 2021 – August 2024

Diploma in Backend Engineering — AltSchool Africa | April 2023 – April 2024

AI Career Essentials Certificate — ALX | March 2024 – May 2024

Backend Engineering Finalist Certificate — HNG Tech | 2025

LEADERSHIP

Google Developer Student Club — Founding Lead · *Arthur Jarvis University* | August 2022 – August 2023

- Founded and led the university's first GDSC chapter, organising 12+ technical workshops on web development, APIs, and cloud technologies.
- Mentored 5+ students in software engineering and web development practice.